

EXAMPLE FROM PREVIOUS LECTURE

Ex: Sampling without replacement

- A president (P) and a treasurer (T) chosen from 20 people including Alice, Bob.
- Probability that Alice is president and Bob is treasurer?

$$\# \text{ of ways to select (P, T): } 20 \times 19 = 380 \quad \Rightarrow \quad \frac{1}{380}$$

IN GENERAL

If ordered selection of k items out of n is done one at a time **without replacement**, there are $n \times (n - 1) \times \cdots \times (n - k + 1) = \frac{n!}{(n - k)!}$ outcomes (k items in order of selection)

FURTHER EXAMPLES

Ex: Birthday problem

- Probability that 2 in a group of 20 have same birthday?

Size of $\Omega = 365^{20}$ (Sampling with replacement)

of outcomes s.t. all birthdays are different $= \frac{365!}{(365 - 20)!}$
(Sampling without replacement)

$$\Rightarrow \text{Probability} = 1 - \frac{365!}{(365 - 20)! \times 365^{20}} = 0.411$$

PROBABILITY IN LIGHT OF NEW INFORMATION

Judgment under Uncertainty: Heuristics and Biases

Biases in judgments reveal some heuristics of
thinking under uncertainty.

Amos Tversky and Daniel Kahneman

Individual described by:

former neighbor as follows: “Steve is very shy and withdrawn, invariably helpful, but with little interest in people, or in the world of reality. A meek and tidy soul, he has a need for order and structure, and a passion for detail.”

Is Steve a librarian or a farmer?

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Is Steve a librarian or a farmer?

- Assume:
 - Ω only farmers (F), librarians (L). $|F| = 100$, $|L| = 5$
 - 40% of L and 10% of F fit the description.

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- Probability a randomly picked person *Farmer* is $20/21$.

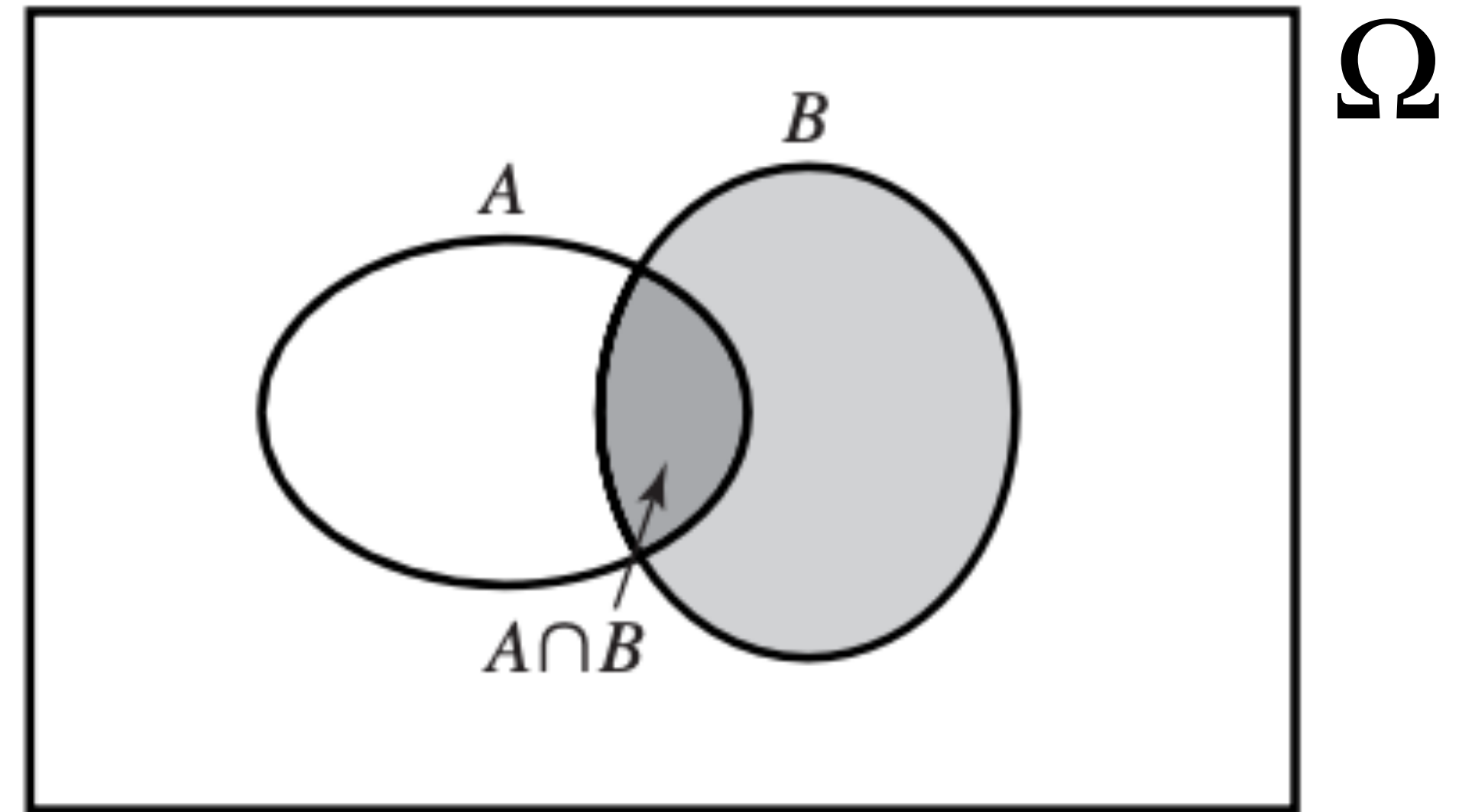
PROBABILITY IN LIGHT OF NEW INFORMATION

- From the opposite direction:
Also a mistake to ignore new information and not update prior probabilities.
- Probability a randomly picked person F is $20/21$.
Say **new information**. We know person picked fits description:

former neighbor as follows: “Steve is very shy and withdrawn, invariably helpful, but with little interest in people, or in the world of reality. A meek and tidy soul, he has a need for order and structure, and a passion for detail.”
- Probability that picked person is F still $20/21$? **No! Decreases.**

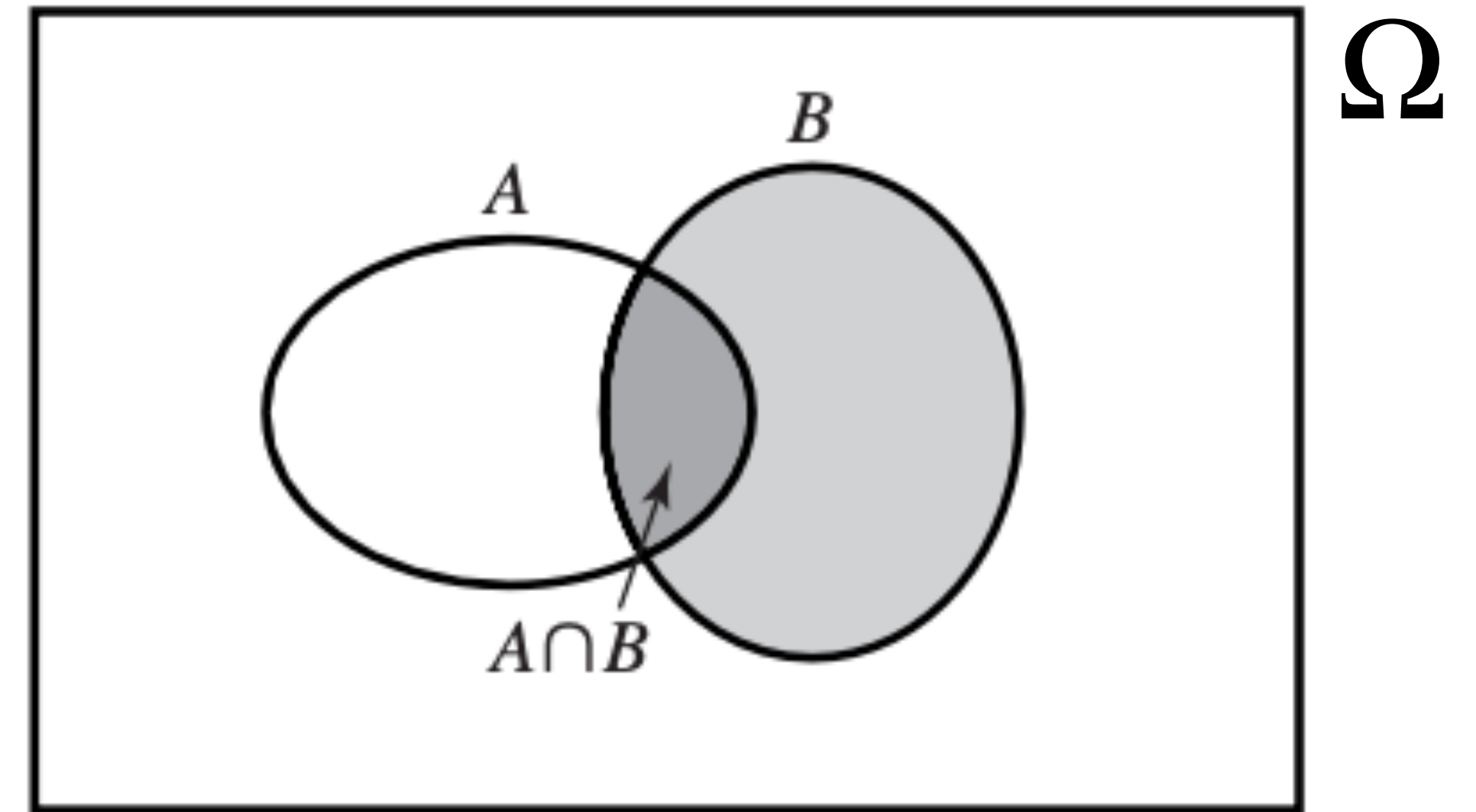
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Interpret with the previous example:

- Ω only farmers (F), librarians (L). $|F| = 100$, $|L| = 5$
- 40% of L and 10% of F fit the description.

People fitting the description: 2 L and 10 F .

$$P(F | D) = \frac{P(F \cap D)}{P(D)} = (10/105)/(12/105) = 10/12$$

CONDITIONAL PROBABILITIES

Ex: Two fair dice rolled.

A : At least one of two dice is even,

B : Sum of two dice is 7 or 11. $P(A | B), P(B | A)?$



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$$B \subset A \Rightarrow P(A \cap B) = P(B) \Rightarrow P(A | B) = P(A \cap B) / P(B) = 1$$

$$P(A) = 1 - P(A^c) = 1 - 9/36 = 27/36, \quad P(B) = 8/36$$

$$P(B | A) = P(A \cap B) / P(A) = (8/36) / (27/36) = 8/27$$

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Probability that $T = 5$ observed before $T = 6$?

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Restrict attention to new space: outcome either $T = 5$ or $T = 6$.

Let A : event that $T = 5$, B : event that either $T = 5$ or $T = 6$.

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{P(B)} = \frac{4/36}{4/36 + 5/36} = \frac{4}{9}$$

MULTIPLICATION PRINCIPLE

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Note that we also have

$$P(A \cap B) = P(B | A)P(A)$$

MULTIPLICATION PRINCIPLE

Ex: 2 balls drawn, without replacement, from box with r red and b blue balls. Probability that 1st ball red and 2nd ball blue?

MULTIPLICATION PRINCIPLE

Ex: 2 balls drawn, **without replacement**, from box with r red and b blue balls. **Probability that 1st ball red and 2nd ball blue?**

Let A : event that 2nd ball blue, B : event that 1st ball red.

$$P(B) = \frac{r}{r+b}, P(A | B) = \frac{b}{r+b-1} \Rightarrow P(A \cap B) = \frac{rb}{(r+b)(r+b-1)}$$

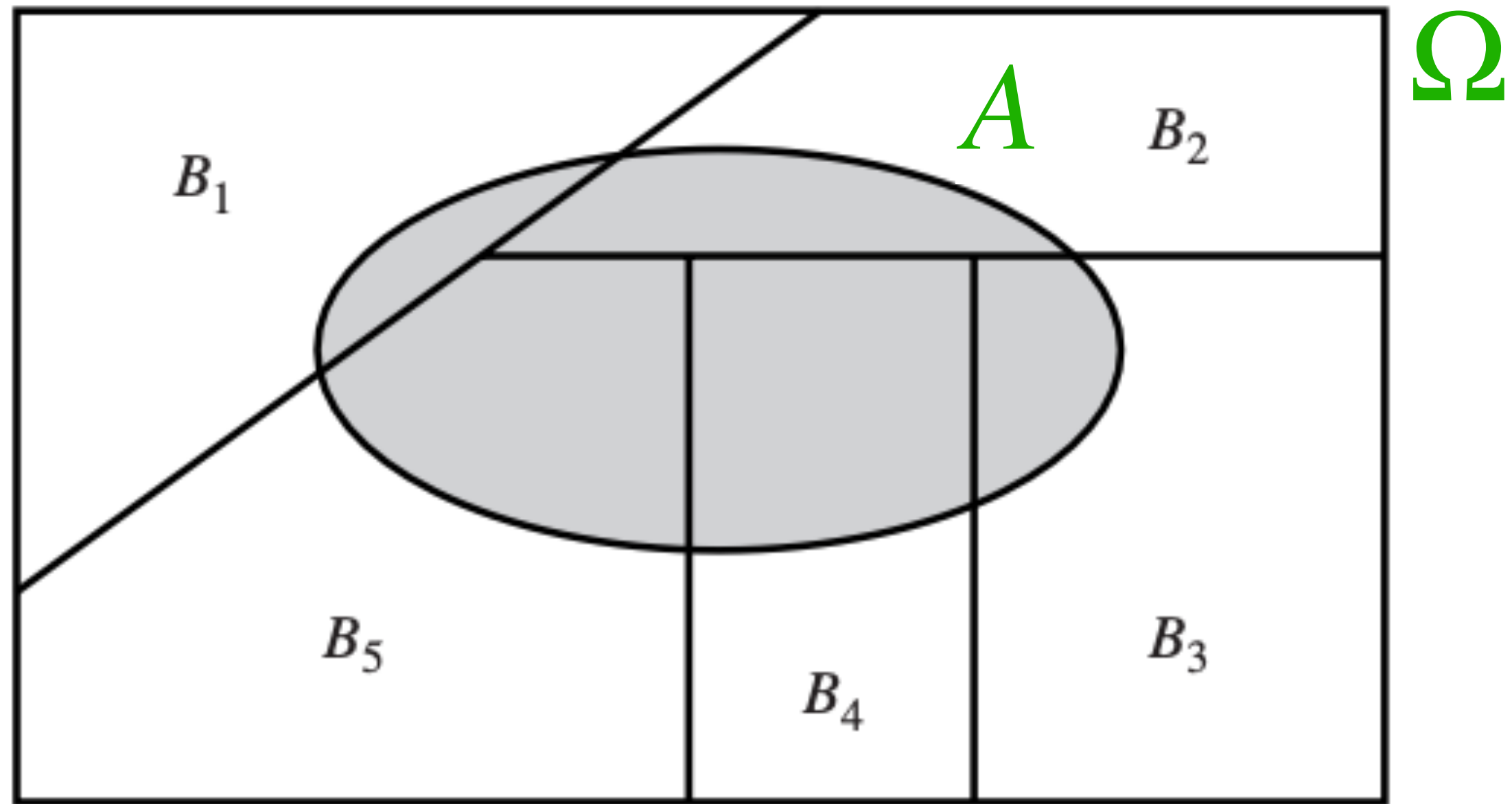
MULTIPLICATION PRINCIPLE

We can also generalize to three or more events. For three:

$$P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B)$$

LAW OF TOTAL PROBABILITY

Partition of sample space makes probability calculations easy.



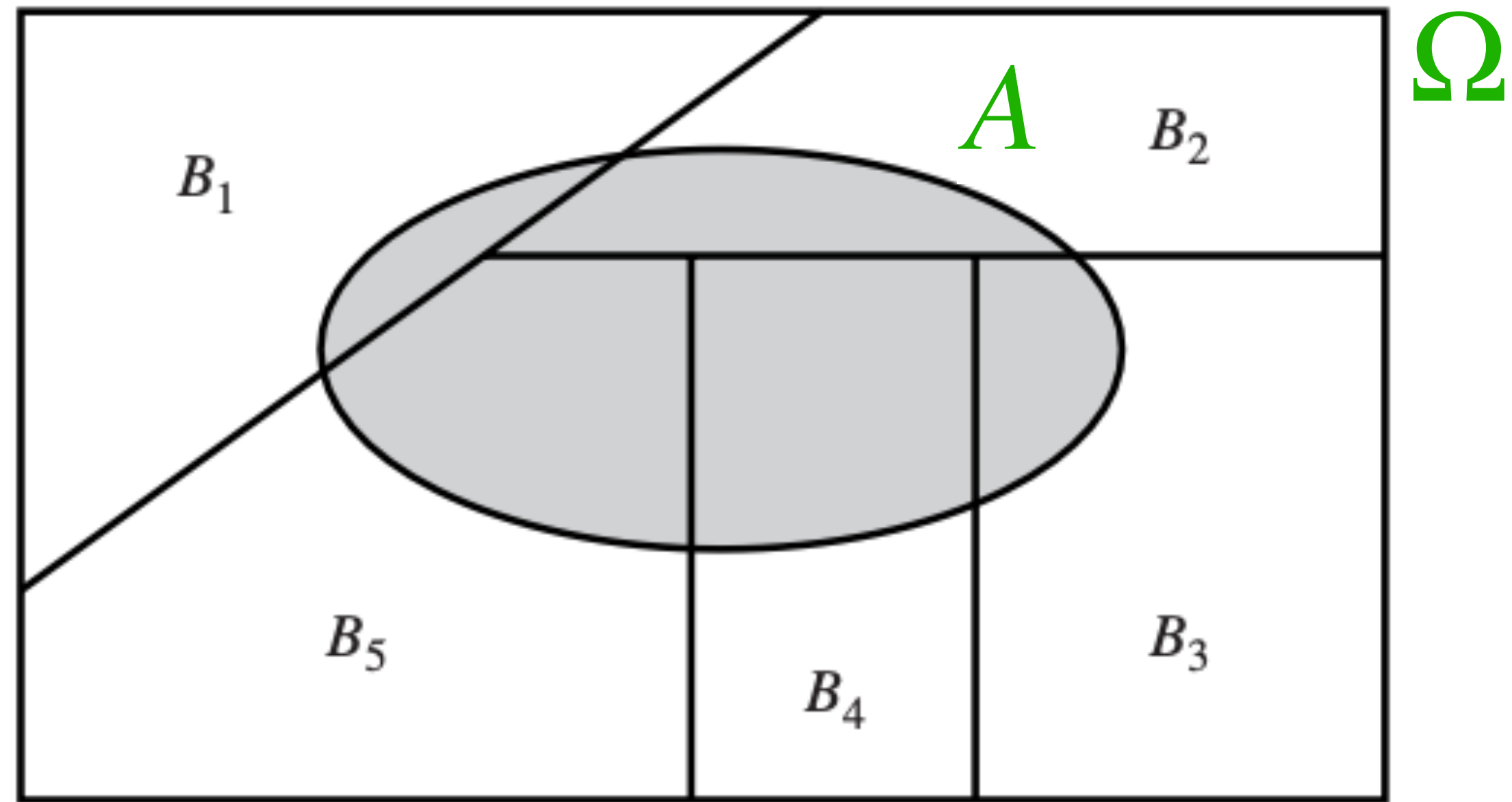
$$i \neq j \Rightarrow B_i \cap B_j = \emptyset$$

AND

$$\bigcup_{\forall i} B_i = \Omega$$

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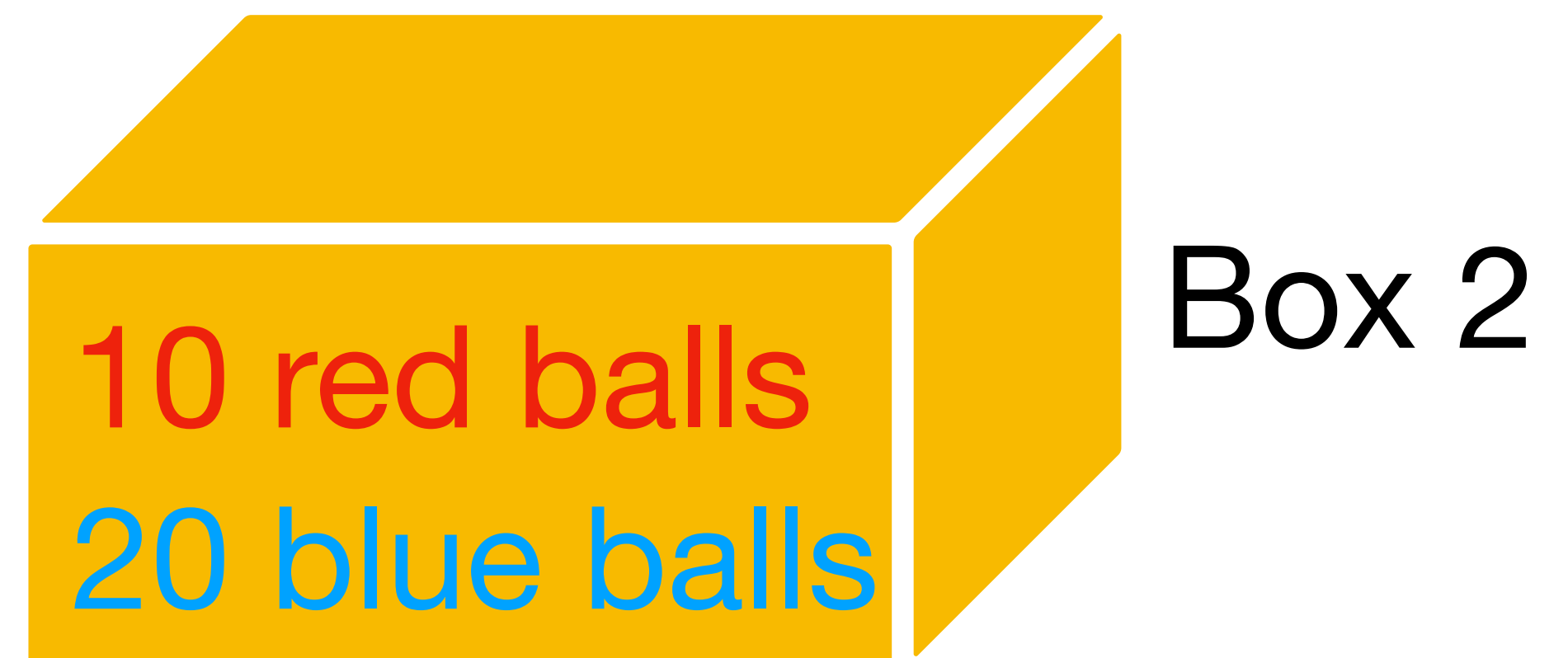
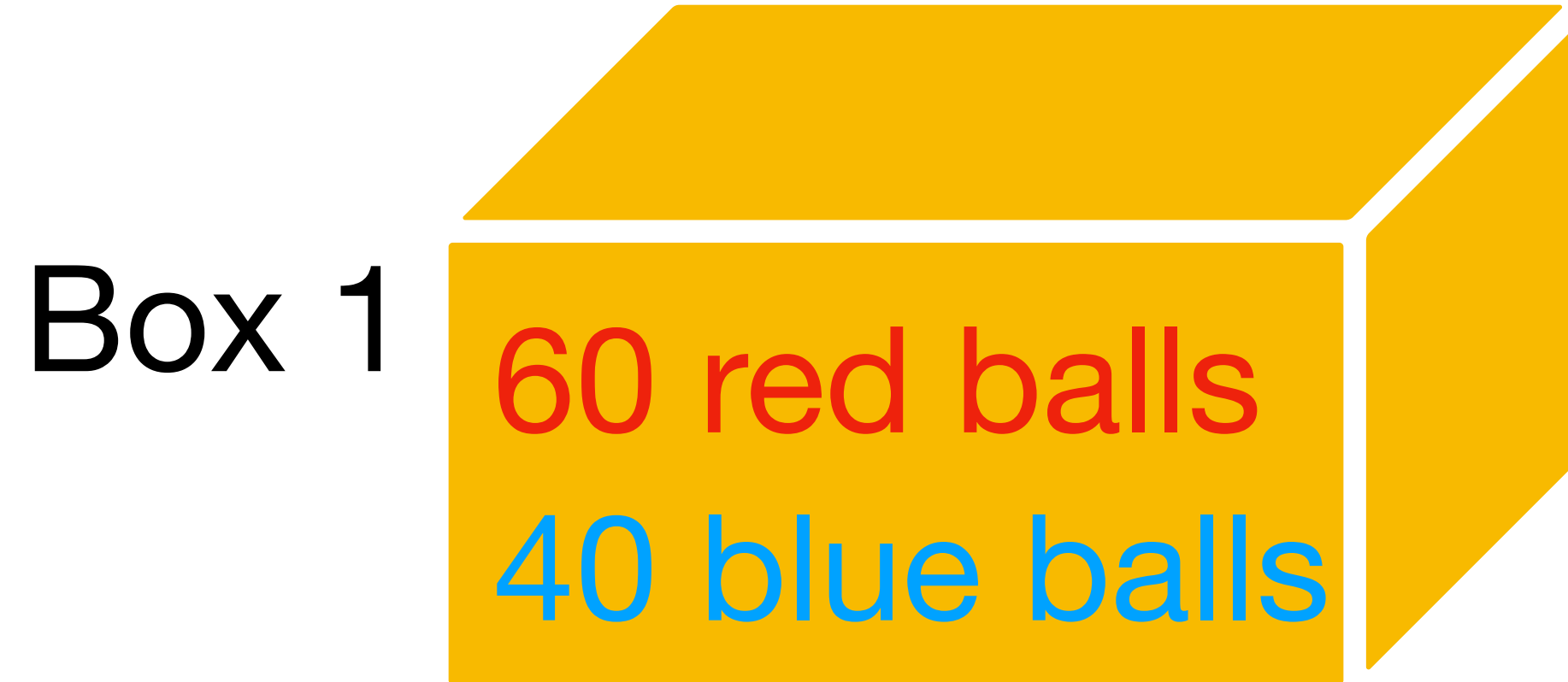
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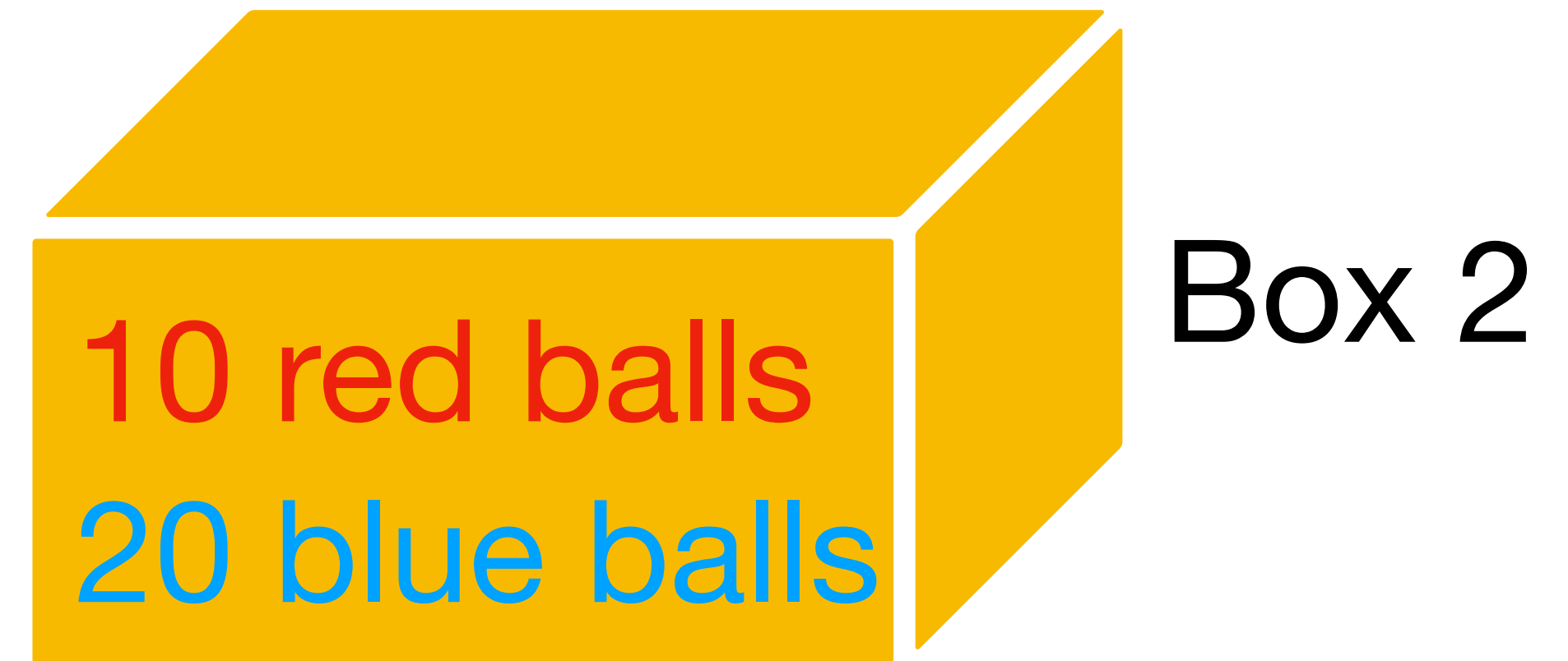
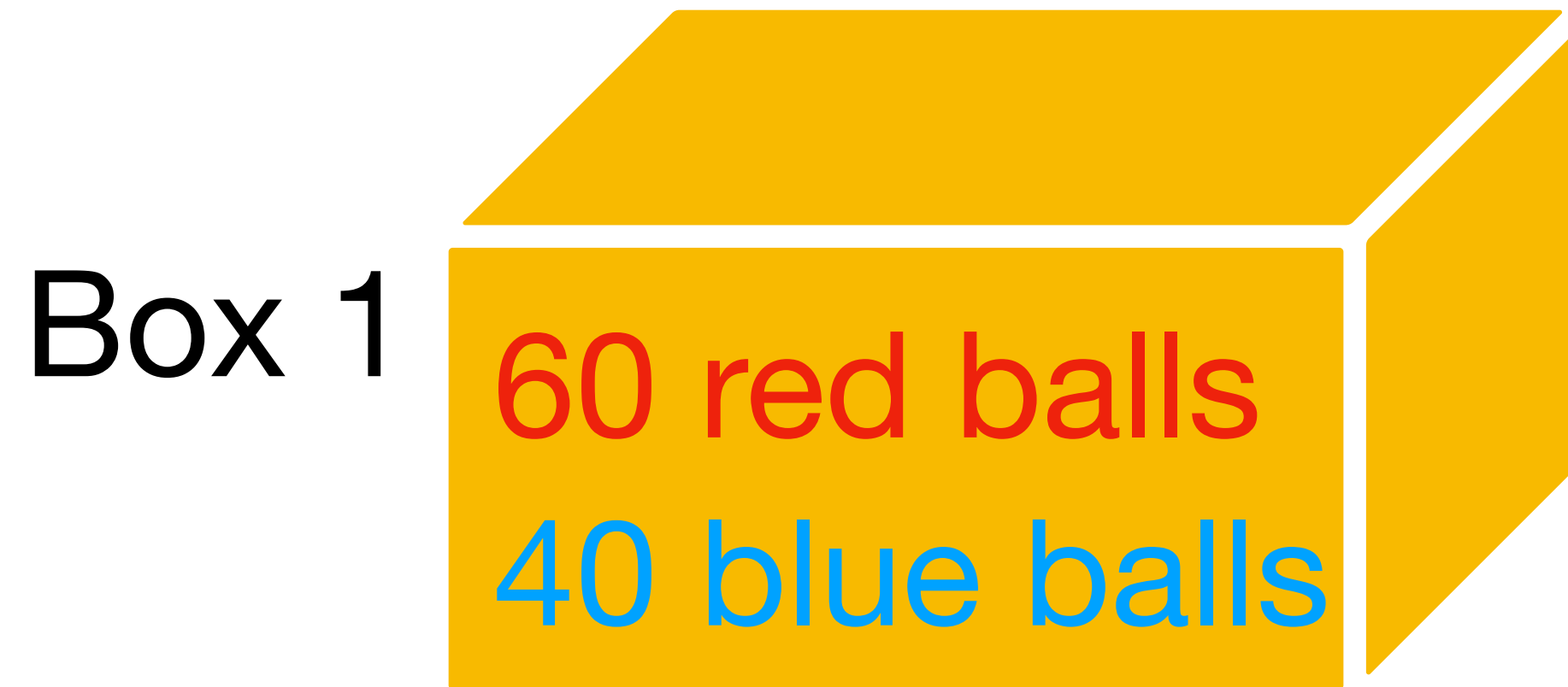
For any event A ,
$$P(A) = \sum_{\forall i} P(B_i)P(A | B_i)$$

LAW OF TOTAL PROBABILITY



Ex: Select a box randomly and select a ball from it randomly.
Probability that the selected ball is red?

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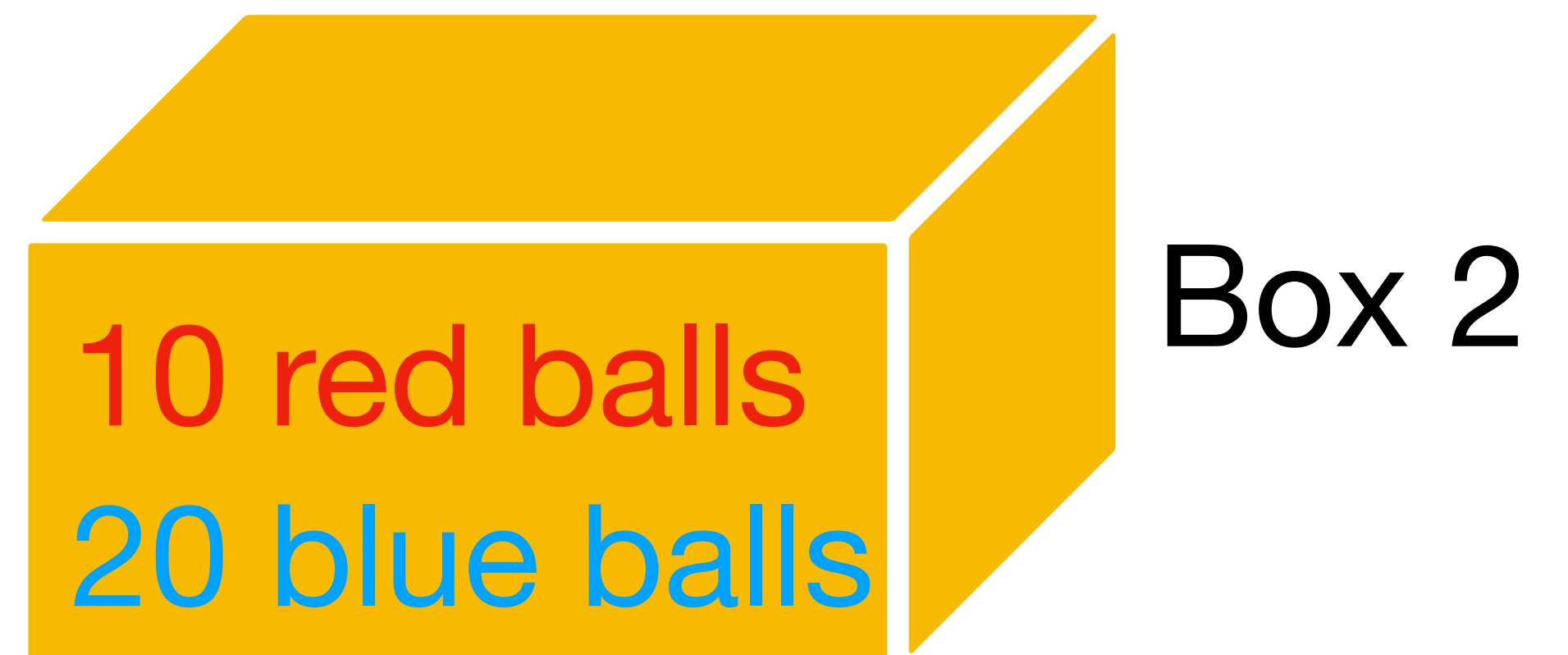
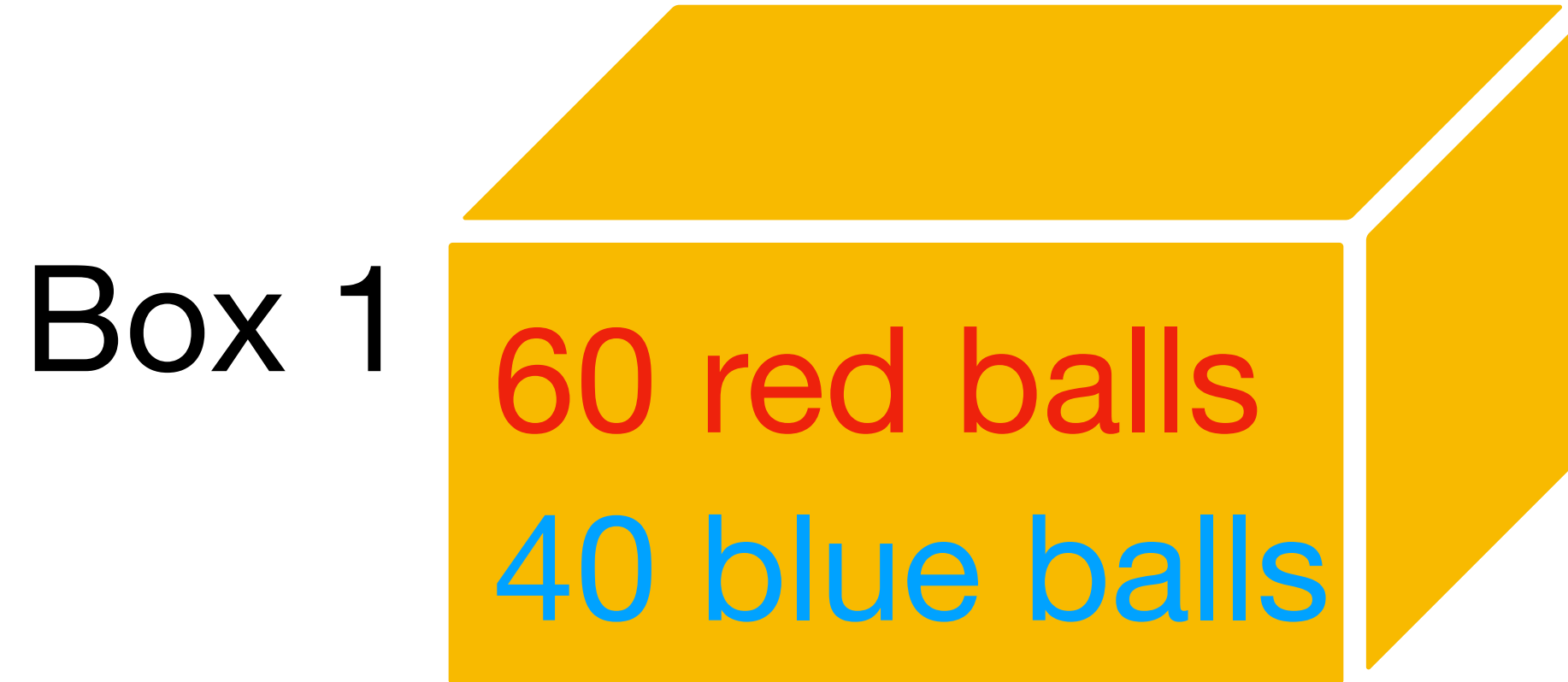


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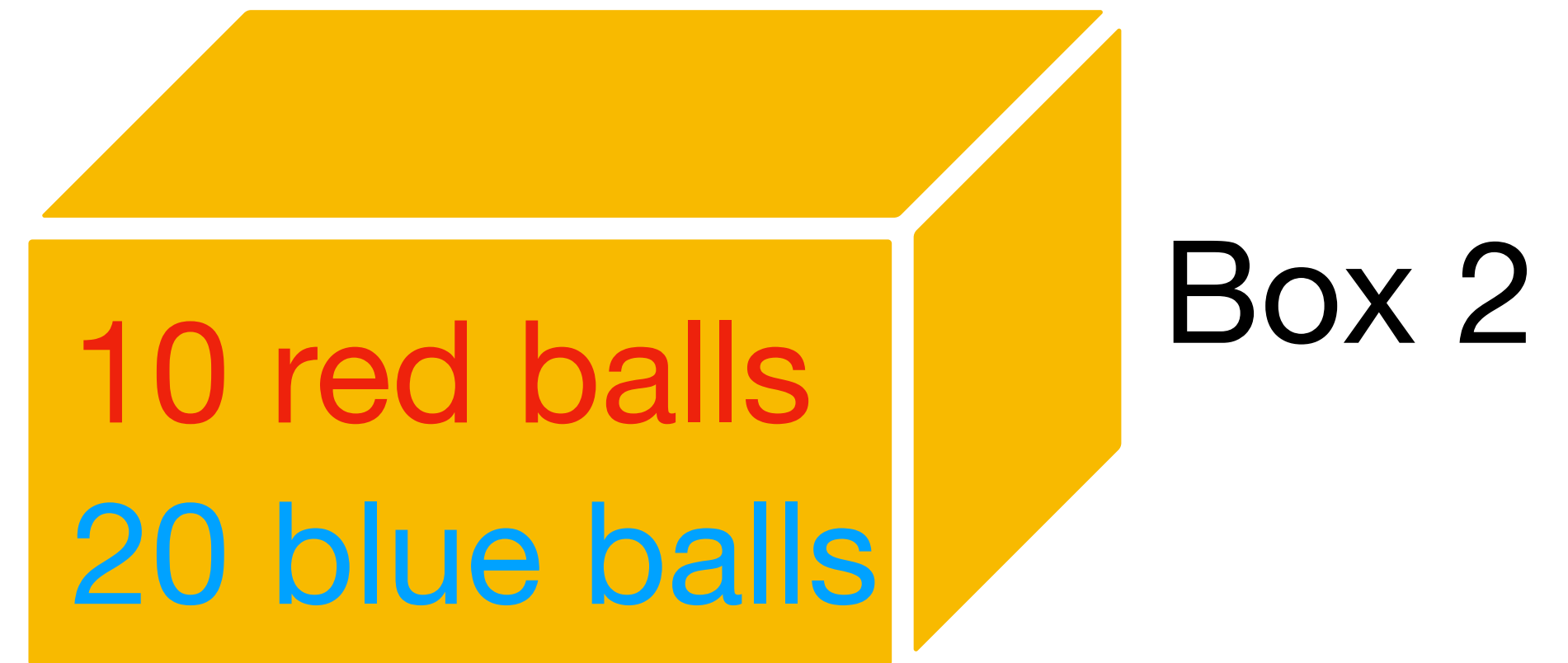
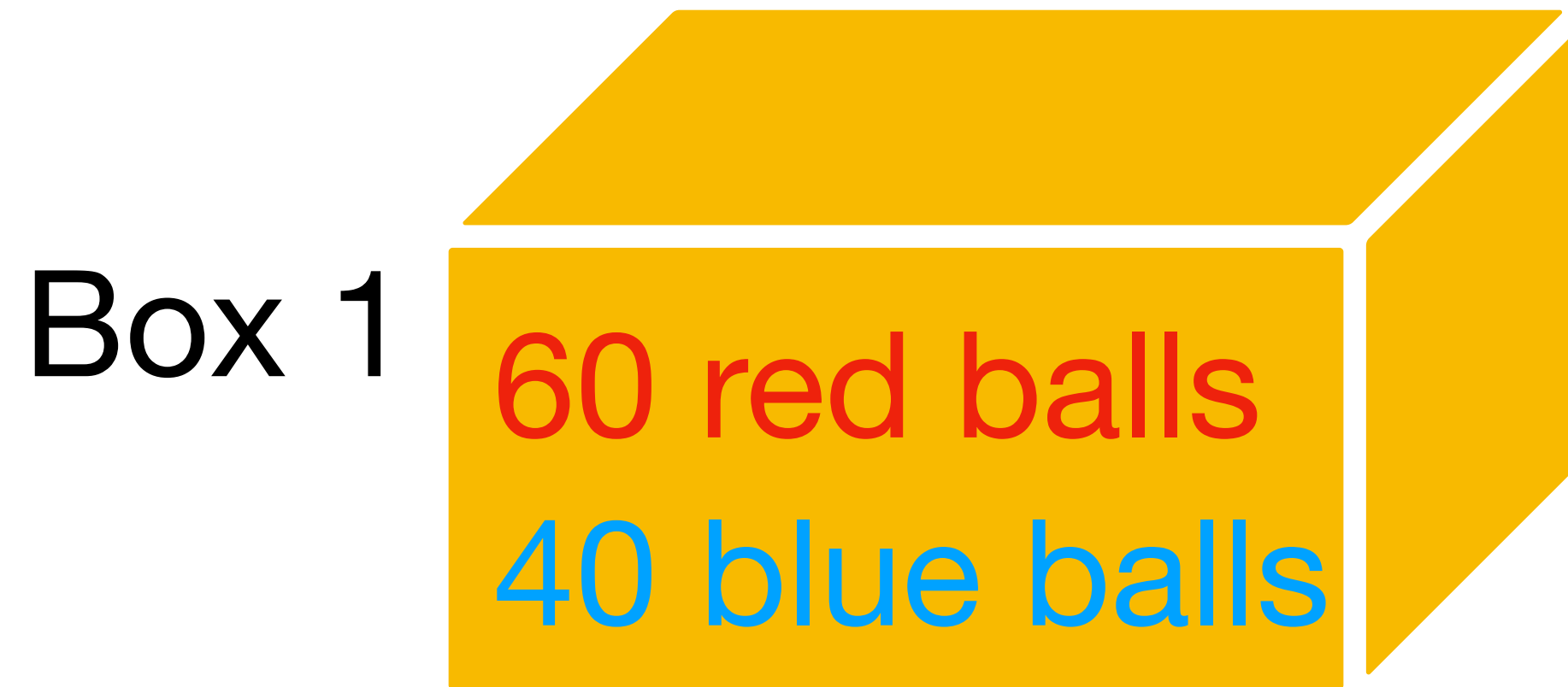
$$P(A) = \sum_{i=1,2} P(B_i)P(A | B_i) = \frac{1}{2} \times \frac{60}{100} + \frac{1}{2} \times \frac{10}{30} = \frac{7}{15}$$

BAYES' THEOREM



Ex: Someone selects a box, selects a ball from it (both random).
We are told it is **red**. **Probability that Box 1 was selected?**

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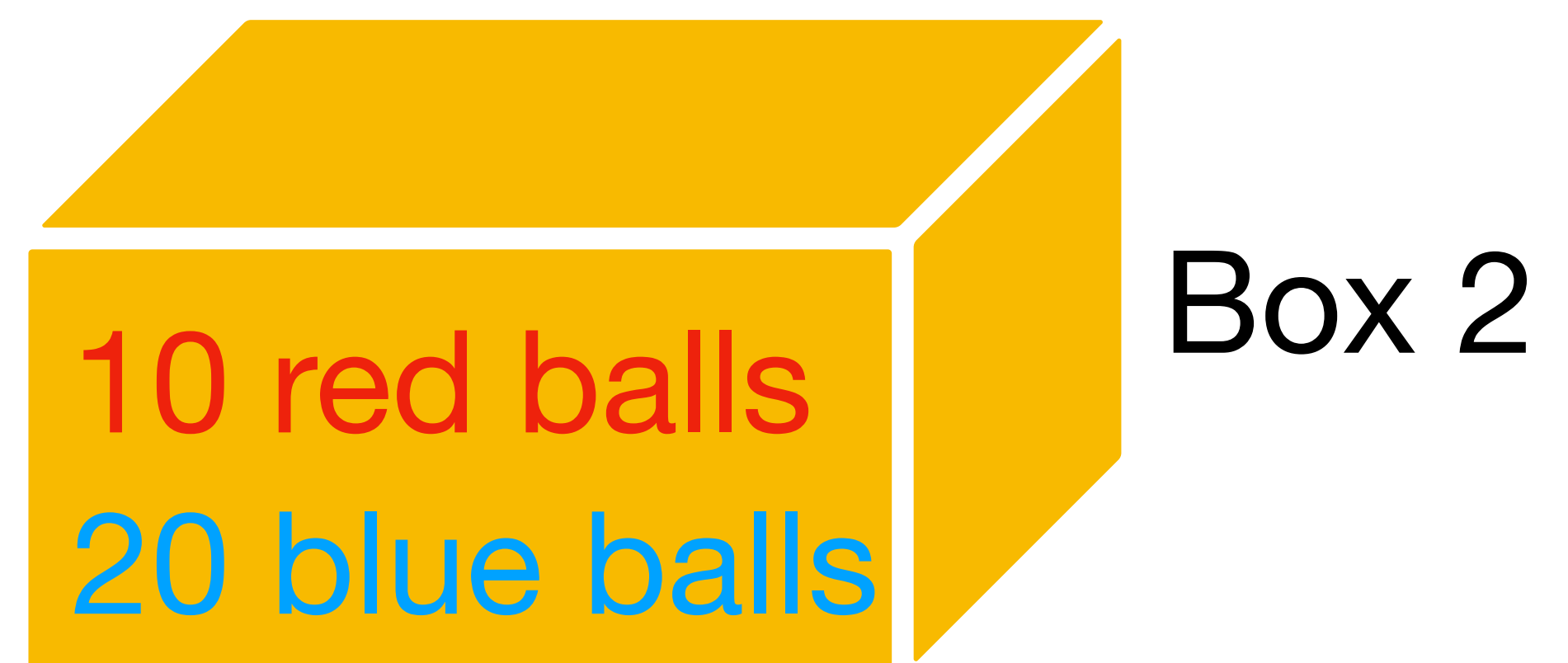
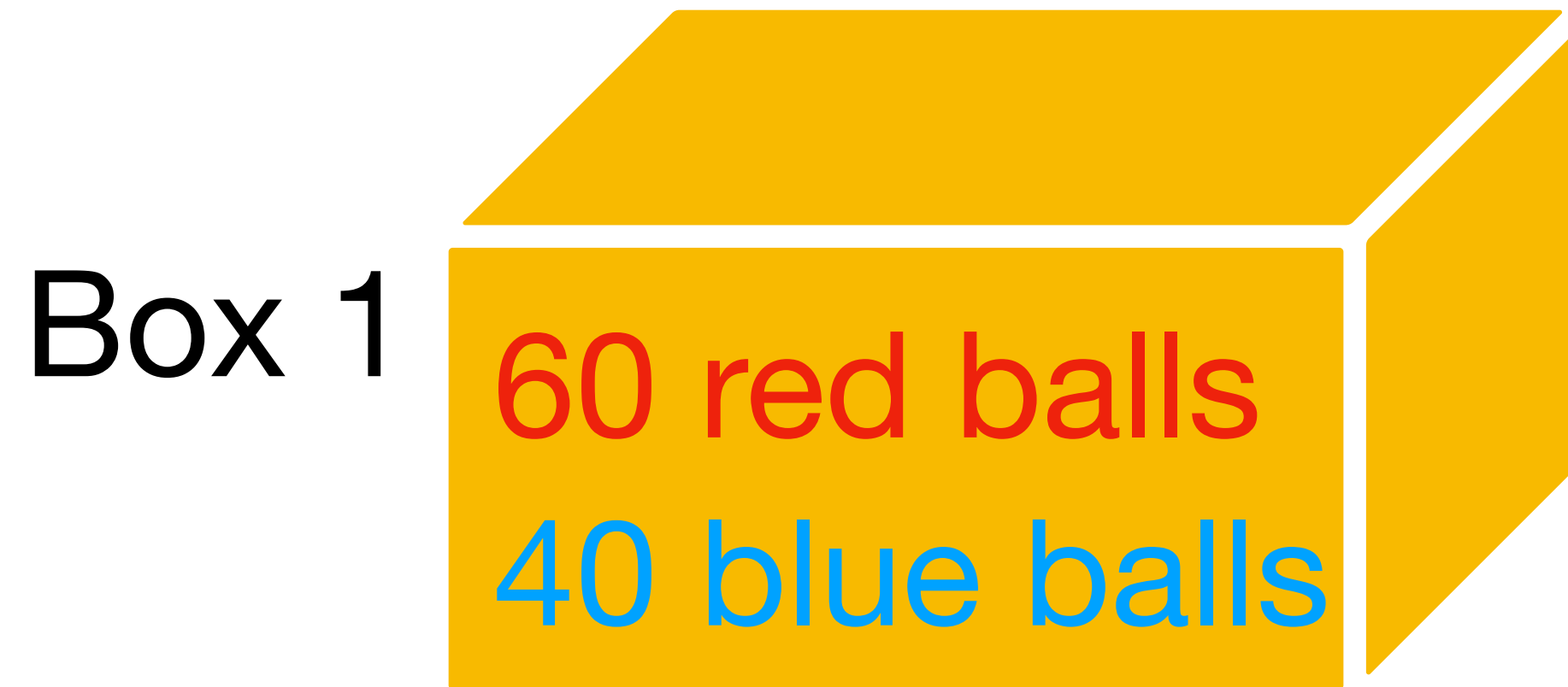


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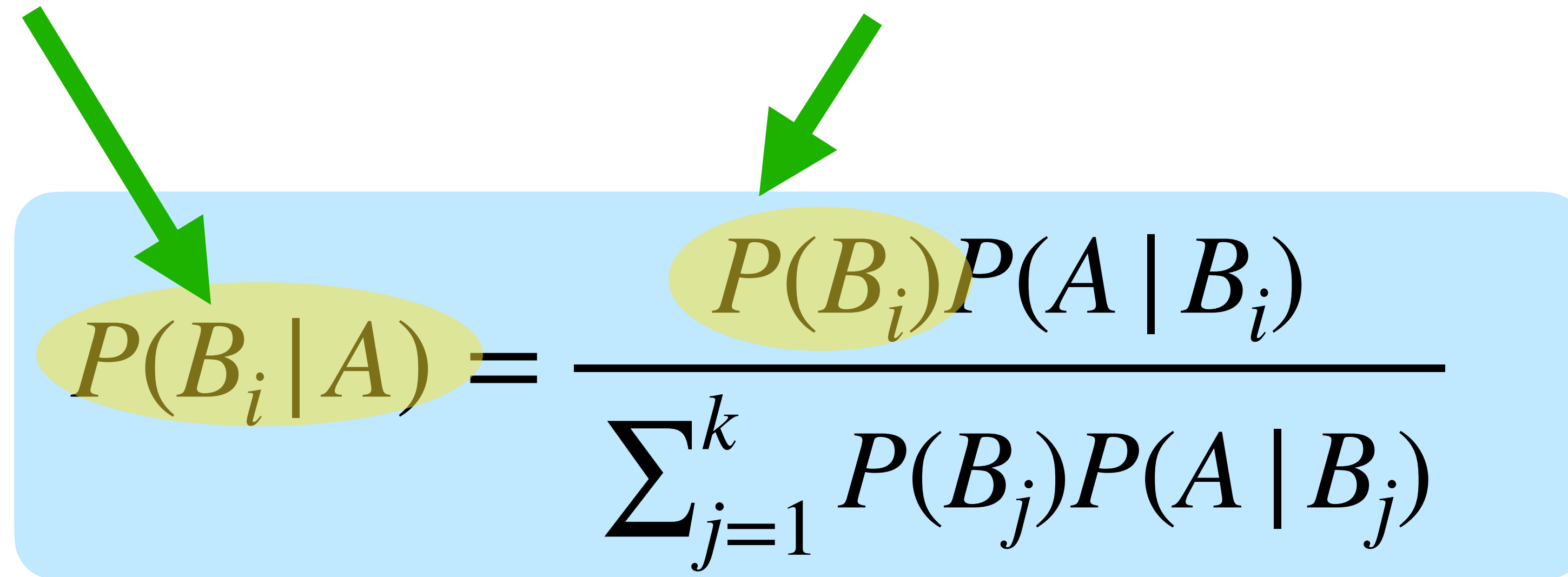
$$P(B_i | A) = \frac{P(B_i)P(A | B_i)}{\sum_{j=1}^k P(B_j)P(A | B_j)}$$

where $B_1, B_2, \dots, B_k$ partition Ω .

BAYES' THEOREM

Posterior probability

Prior probability



The diagram shows the Bayes' Theorem formula: $P(B_i | A) = \frac{P(B_i)P(A | B_i)}{\sum_{j=1}^k P(B_j)P(A | B_j)}$. The entire formula is enclosed in a light blue rounded rectangle. The term $P(B_i | A)$ on the left is highlighted with a yellow oval, and a green arrow points from the text 'Posterior probability' above to it. The term $P(B_i)$ in the numerator is also highlighted with a yellow oval, and a green arrow points from the text 'Prior probability' above to it.

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Updates prior probability of B_i
based on information gathered from A